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Palaeomagnetism of Precambrian rocks from Gabon, Congo craton, Africa

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Abstract - Palaeomagnetic results for 75 orientated hand samples (25 sampling sites) from the Ogooué Series amphibolites (27), Archean basement amphibolites and granulites (14) and Precambrian dolerite dykes (34) are presented. Magnetization of many amphibolites showed unstable directions during alternating field (AF) and thermal demagnetization experiments and no directions of magnetization could be isolated. However, a characteristic direction could be isolated in a few amphibolite sites and for the granulites studied. The best group of site mean directions Dm=54°, Im=69° (κ=27, α_{95}=18°, N=4) yielded a palaeomagnetic pole located at 44° E, 19° N (K=10, A_{95}=30°). This pole is close to a palaeomagnetic pole obtained for granulites from the Jequié complex (São Francisco craton) dated by the $^{40}Ar/^{39}Ar$ method as 2.0 Ga, suggesting that both areas belonged to the same tectonic unit at that time. The unmetamorphosed dolerite dykes (5) displayed multi-component behavior, but a common characteristic direction could be isolated for four dykes, Dm=262°, Im=-49° (κ=13, α_{95}=27°, N=4), which yielded a palaeomagnetic pole located at 72° E, 8° N (K=9; A_{95}=33°). However, this pole is different from those obtained for dyke swarms in Brazil, which are supposedly of comparable age.

Résumé - Les résultats paléomagnétiques de 75 échantillons orientés (25 sites) sont présentés. Ils concernent les ensembles lithologiques suivants: amphibolites de la Série de l'Ogooué (27), amphibolites et granulites d'âge archéen du complexe cristallin (14) et dolérites provenant de dykes Précambrien (34). L'aimantation de la majorité des amphibolites montrent des directions instables lors du traitement des champs alternatifs et thermiques, et aucune direction d'aimantation n'a pu être définie. Cependant, une direction caractéristique pour quelques sites d'amphibolites et pour l'ensemble des granulites a pu être mise en évidence. Concernant le groupe de directions moyennes composé de deux sites de granulites et de deux sites d'amphibolites, Dm=54°; Im=69° (κ=27, α_{95}=18°, N=4), nous avons calculé un pôle paléomagnétique localisé en 44°E; 19°N (K=10; A_{95}=30°). Ce pôle est proche de celui déterminé pour le complexe Jequié (caton de São Francisco) daté à 2,0 Ga par la méthode $^{40}Ar/^{39}Ar$. Pour ces deux zones, cela suggère l'appartenance à la même unité tectonique à cette période. Les dykes de dolérite non-métamorphiques (5) indiquent des directions multiples; cependant, quatre dykes montrent une direction caractéristique commune, Dm=262°; Im=-49° (κ=13; α_{95}=27°, N=4). Il est alors possible de calculer pour ces dykes un pôle paléomagnétique situé en 72°E; 8°N (K=9; α_{95}=33°). Néanmoins, ce pôle est différent de ceux déterminés pour de très nombreux dykes du Brésil considéré comme étant de même âge.

INTRODUCTION

The present Palaeoproterozoic palaeomagnetic data from the Kalahari Craton (southern Africa) enabled the definition of a fairly well-established apparent polar wander (APW) path during the time interval between 2.05 Ga and 1.85 Ga for this cratonic unit (Morgan, 1985; Hatting, 1989).

Several well dated palaeomagnetic poles from the Amazonian and West African cratons in the 2.05-1.8 Ga time interval led Onstott and Dorbor (1987) and Renne *et al.* (1988) to suggest that these cratonic units were separated from the Kalahari craton at that time.

However, only a few palaeomagnetic poles of Palaeoproterozoic age are presently available for the intervening São Francisco/Congo cratonic area.

Unmetamorphosed dykes of Meso- to Neoproterozoic age from the eastern São Francisco craton (Salvador and Ilhéus-Olivença dykes), yielded compatible geochemical, geochronological ($^{40}Ar/^{39}Ar$) and palaeomagnetic results defining an APW path for the São Francisco craton during the interval 1.08-1.01 Ga (d'Agrella-Filho *et al.*, 1989, 1990).

Mafic dykes outcropping in Gabon (Africa) have been correlated with the Salvador and Ilhéus-Olivença dyke swarms, on a pre-drift configuration, based on geological and geochronological evidence (Renne *et al.*, 1990). According to these authors, the Mayumbian basin of the Congo craton, where a 4-5 km thick sequence of basalts and rhyolites has been deposited, could be the magma feeder source for the Brazilian dykes and also the mafic dykes of similar

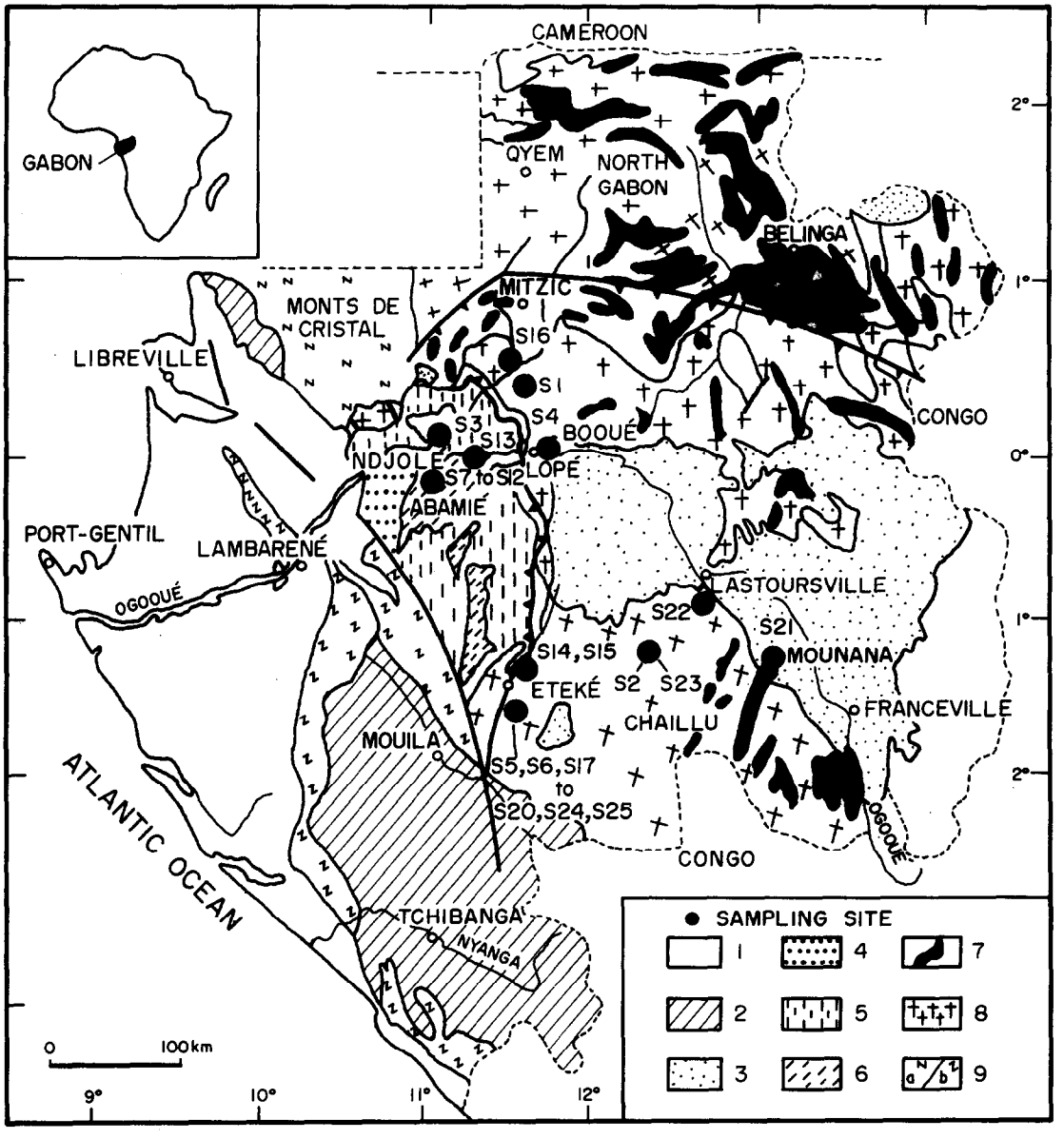


Figure 1. Locations of the sampling sites (full circles). Geological map following Ledru *et al.*, 1989. 1: Mesozoic cover; 2: Neoproterozoic of the Nyanga syncline; 3: Palaeoproterozoic: Francevillian rocks; 4: N'Djolé basin; 5: Ogooué metamorphics; 6: Archaean migmatitic domes; 7: Fe formation; 8: Chaillu and North Gabon granitic domains; 9a: medium- to high-grade gneisses of the Monts de Cristal; 9b: Lambaréné migmatites.

age (1.0 Ga) cutting the Francevillean Group in Gabon. This magma source would geographically agree with the northern north-south striking swarm at Salvador, the western east-west striking swarm at Ilhéus-Olivença and the northeastern east-northeast striking swarm at Gabon. Also, recent geochemical analysis of the Gabon dykes shows that they are similar to the Brazilian dykes (Enzo Piccirillo, *pers. comm.*, 1994).

The main objective of this study was to obtain palaeomagnetic results from selected rock formations on the Congo craton and compare them with those for rocks of similar age from the São Francisco craton. These results can then be used for paleogeographic reconstructions of these cratonic areas and may help to elucidate the evolution of the Palaeoproterozoic Gabon mobile belt, previously interpreted as a suture zone (Ledru *et al.*, 1989). For this purpose, it is important to compare palaeomagnetic results from the cratonic areas from both sides of the mobile belt, before, during and after its evolution.

GEOLOGICAL SETTING AND SAMPLING

The Precambrian of Gabon is divided into three units; the Archean basement, the Palaeoproterozoic and the Neoproterozoic (Fig. 1).

The Archean shield area consists mainly of polyphase medium- to high-grade banded gneisses and greenstone belts with Fe bearing formations, metamorphosed at 3.15 Ga and intruded by granitoids at around 3.0-2.6 Ga.

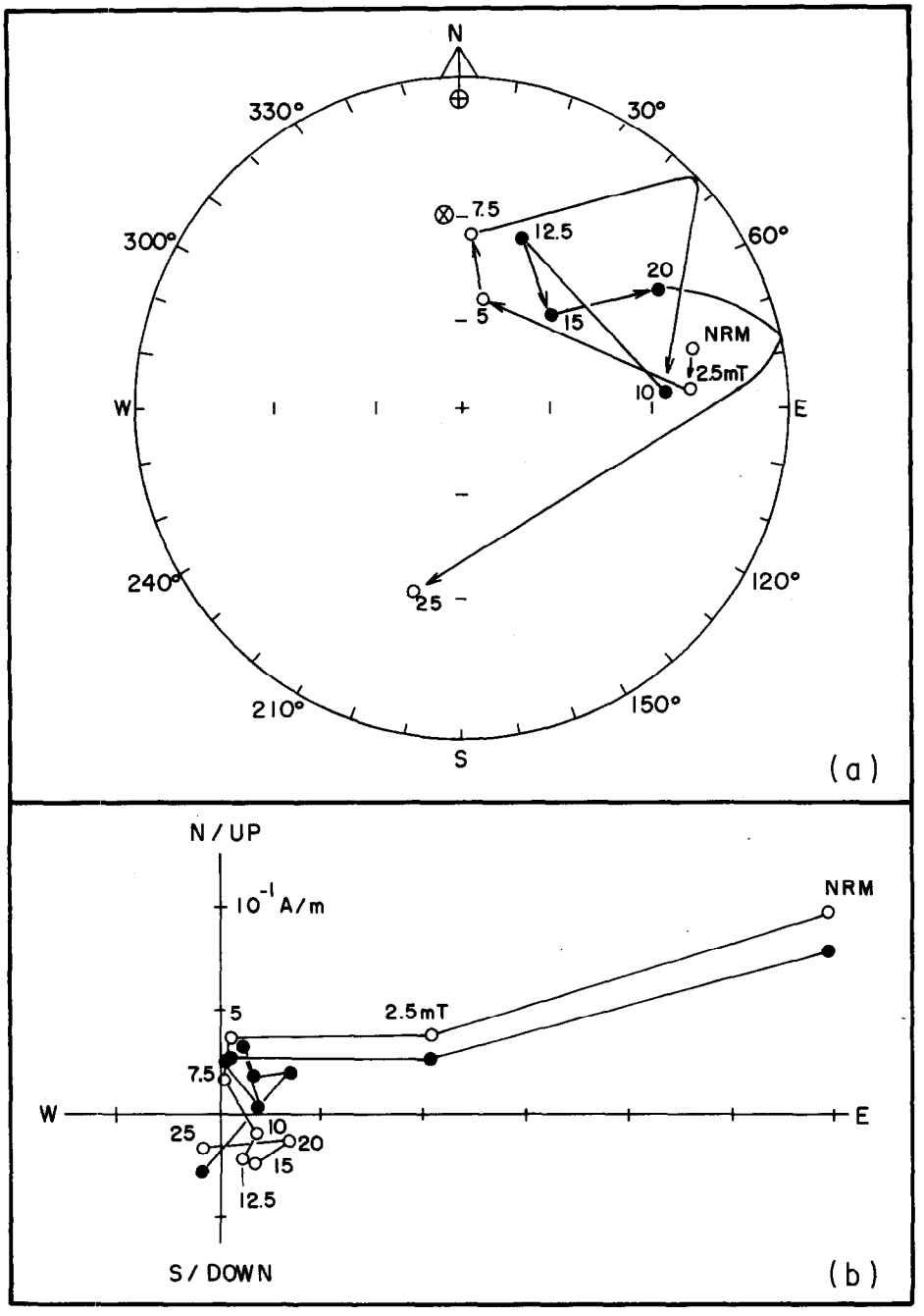


Figure 2. (a) Wulff stereographic projection [open (full) symbols represent upward (downward) inclinations] and (b) Ziljerveld diagram [open (full) circles represent vertical (horizontal) plane projections] of magnetization directions during treatment for sample OG-4-A1, site S8. ⊗ and ⊕ represent the present geomagnetic field and the present dipolar field, respectively.

The Palaeoproterozoic is represented by the Francevillian sedimentary rocks (2.3-2.0 Ga) and the Ogooué fold and thrust belt. Resting unconformably over the Archean basement, the Francevillian is a dominantly clastic series of uraniferous conglomerates and sandstones, arkosic sandstones with layer intervals of cherts and manganese oxides and, locally, volcanic deposits.

The Ogooué orogenic belt consists mainly of metasediments and volcanic rocks considered to be Palaeoproterozoic in age. Locally, slices of reworked Archean rocks have been tectonically incorporated. The metamorphic and structural features are consistent with the collision tectonics of the Eburnean cycle (2.2-1.9 Ga).

The Upper Proterozoic consists mainly of the Nyanga and Mayombe sediments. To the west, the Mayombe Pan-African belt is composed of Neoproterozoic sediments and reworked Archean and Palaeoproterozoic basement rocks that overthrust the Nyanga Series.

Several dolerite dykes intrude the Lower Proterozoic

Table 1. Summary of the palaeomagnetic results

Site	Samples	Lithology	Geological formation	NRM E-4 (A m^{-1})	X E-4 (SI)	N (n)	Mean magnetic direction					Palaeomagnetic pole	
							Dec(°)	Inc(°)	K	α_{95}(°)	R	PLong(°E)	PLat(°E)
S1	59-61	Granulite	Archaean	10,072	1108	3(9)	34	62.0	24	26	2.916075	42.4	37.5
S2	74-75	Granulite	Archaean	1796	838	2(7)	95.4	67.6	398	13	1.997449	52.8	-4.4
S3	53-55	Amphibole	Archaean	2	8	-	-	-	-	-	-	-	-
S4	62-64	Amphibole	Archaean	0.1	5	1(3)	268.5	-49.6	196	9	2.989781	251.3	-1.3
S5	79-80	Amphibole	Archaean	9	10	2(5)	305.3	50.2	123	23	1.991838	318.6	28.8
S6	93	Amphibole	Archaean	44	110	-	-	-	-	-	-	-	-
S7	1-3	Amphibole	Ogooué S.	359	127	-	-	-	-	-	-	-	-
S8	4-6	Amphibole	Ogooué S.	174	12	-	-	-	-	-	-	-	-
S9	7-9	Amphibole	Ogooué S.	291	10	2(6)	111.5	-56.0	77	29	1.987095	139.6	-17.1
S10	10-12	Amphibole	Ogooué S.	4	8	-	-	-	-	-	-	-	-
S11	13-15	Amphibole	Ogooué S.	1890	11	2(5)	30.4	57.4	672	10	1.998511	43.9	42.7
S12	16-18	Amphibole	Ogooué S.	79	37	-	-	-	-	-	-	-	-
S13	50-52	Amphibole	Ogooué S.	0.1	9	-	-	-	-	-	-	-	-
S14	99-101	Amphibole	Ogooué S.	6132	4715	-	-	-	-	-	-	-	-
S15	102-104	Amphibole	Ogooué S.	39	47	2(6)	85.7	77.1	272	15	1.996323	36.1	0.6
S16	56-58	Amphibole	Met. dyke	1	10	1(4)	153.0	-65.4	98	9	3.969247	168.8	-37.5
S17	81-83	Amphibole	Met. dyke	0.8	8	-	-	-	-	-	-	-	-
S18	84-86	Amphibole	Met. dyke	0.7	12	2(8)	288.6	-24.6	404	13	1.997524	268.5	18.5
S19	95-98	Amphibole	Met. dyke	131	14	-	-	-	-	-	-	-	-
S20	105-107	Amphibole	Met. dyke	15	8	2(6)	150.7	63.4	27	50	1.963111	38.3	-39.6
S21	65-70	Dolerite	Dyke	1377	387	6(20)	287.3	-48.7	452	3	5.988933	252.6	15.6
S22	71-73	Dolerite	Dyke	11,172	315	3(12)	98.6	29.6	246	8	2.991859	87.8	-8.5
S23	76-78	Dolerite	Dyke	2219	181	2(6)	41.1	58.7	385	13	1.997402	51.5	34.6
S24	87-89	Dolerite	Dyke	6513	290	-	-	-	-	-	-	-	-
S25	90-92	Dolerite	Dyke	883	275	2(8)	246.2	-46.8	152	20	1.993436	250.8	-20.0

NRM: mean natural remnant magnetization; X: mean susceptibility; N(n): number of samples used for site mean direction (number of specimens used in each site); K: Fisher precision parameter; α_{95}: Fisher semi-circular cone of confidence; PLong: palaeolongitude; PLat: palaeolatitude.

For sites with N=1, the indicated Fisher's statisitcal parameters are those obtained for sample mean directions.

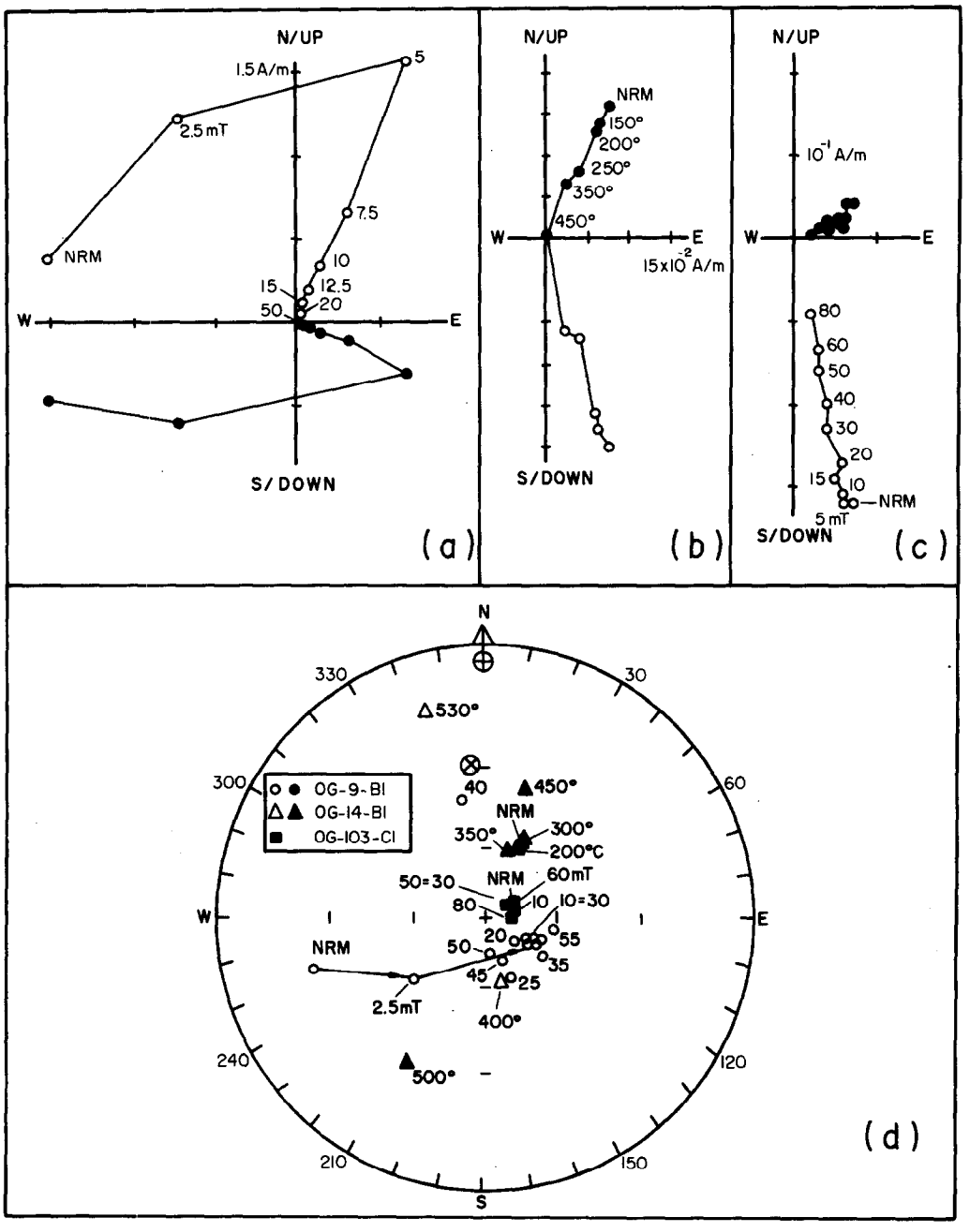


Figure 3. Zijderveld diagrams for samples (a) OG-9-B, site S9; (b) OG-14-B1, site S11 and (c) OG-103-C1, site S15 (Ogooué amphibolites). (d) The respective wulff stereographic projections. Same conventions as for Fig. 2.

and Archean terrains. A dolerite dyke from Mikouloungu (Gabon) has been dated by the K-Ar method. Three concordant results, one on whole rock and two on feldspathic fractions, yielded ages between 995±15 Ma and 977±14 Ma (reference in Cahen *et al.*, 1984).

Field work was carried out in Gabon in August, 1990. Seventy five orientated hand samples from 25 sites were collected for palaeomagnetic analysis (Fig. 1). Twenty seven samples (9 sites) are from amphibolites from the Ogooué Series. Nine amphibolite samples (4 sites) and 5 granulite samples (two sites) are from the Archean basement. Thirty four samples (ten sites) are from metamorphosed and unmetamorphosed dykes. Whenever possible, both solar and magnetic compasses were used for orientating the samples.

PROCEDURE

Samples were treated using standard palaeomagnetic procedures. Three to four cylindrical specimens with diameter and height equal to 2.5 cm were cut from cores drilled from every hand sample. Alternating field (AF) and thermal demagnetization experiments were applied to isolate magnetization components and to test the stability of remnance.

Generally, steps of 2.5, 5 or 10 mT up to 100 mT

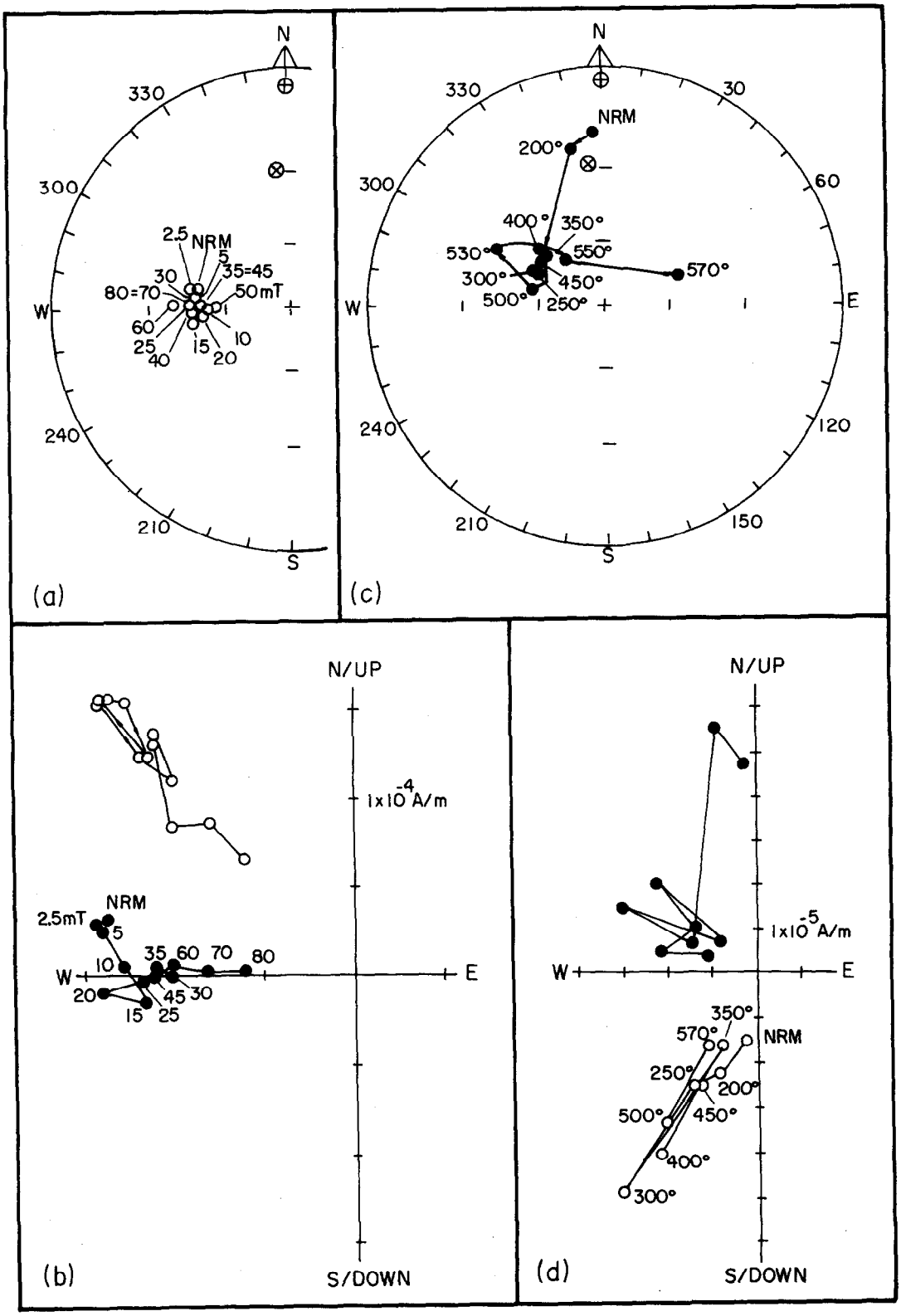


Figure 4. Wulff stereographic projection and Zijderveld diagrams of magnetization directions during treatment for samples (a, b) OG-63-A1, site S4 and (c, d) OG-79-B1, site S5 (Archean amphibolites). Same conventions as for Fig. 2.

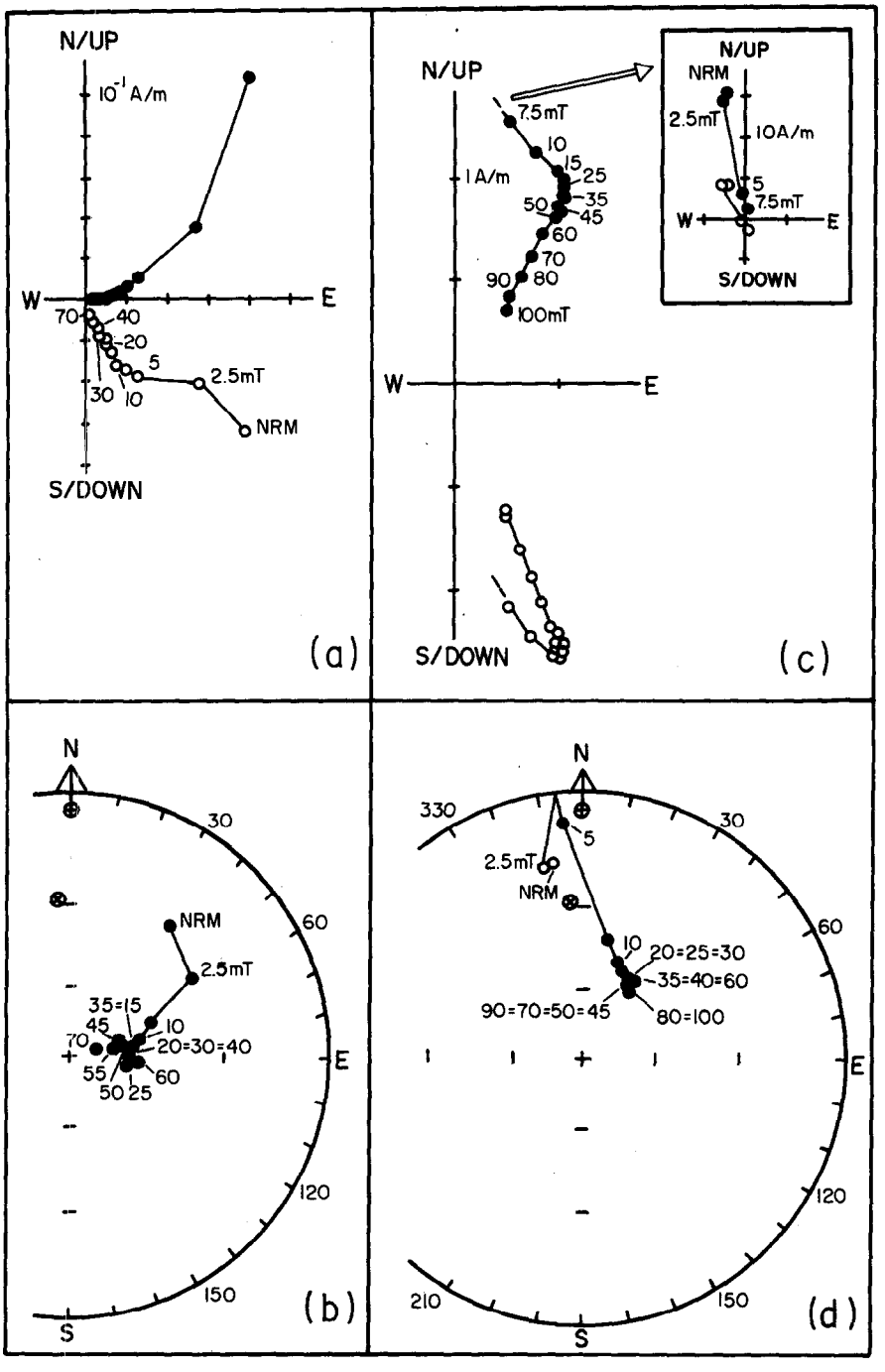


Figure 5. Zijderveld diagrams and wulff stereographic projection for samples (a, b) OG-75-D1, site S2 and (c, d) OG-59-A1, site S1 (Archean granulites). Same conventions as for Fig. 2.

were used for AF treatment and steps of 50°C up to 500°C followed by the sequence 530°C, 550°C, 570°C, 600°C, 630°C, 650°C and 670°C (or 680°C) were used for thermal treatment. This procedure was interrupted whenever the sample remnant magnetization behaved unstably. Sometimes a combination of AF followed by thermal demagnetization was applied (Roy and Lapointe, 1978).

A spinner magnetometer (Molspin) and a susceptibility meter (Sapphire instruments) were used for measurements of the remnant magnetization and magnetic susceptibility. Thermomagnetic curves were obtained for several samples using a horizontal magnetic balance.

Stereographic (Wulff) and orthogonal (Zijderveld, 1967) projections were used to separate magnetizations present in rocks and the principal component analysis was used to determine directions (Kirshvink, 1980). Only directions defined by at least 4 points and with MAD smaller than 10° were considered. Fisher (1953) statistics was used to determine mean directions and palaeomagnetic poles.

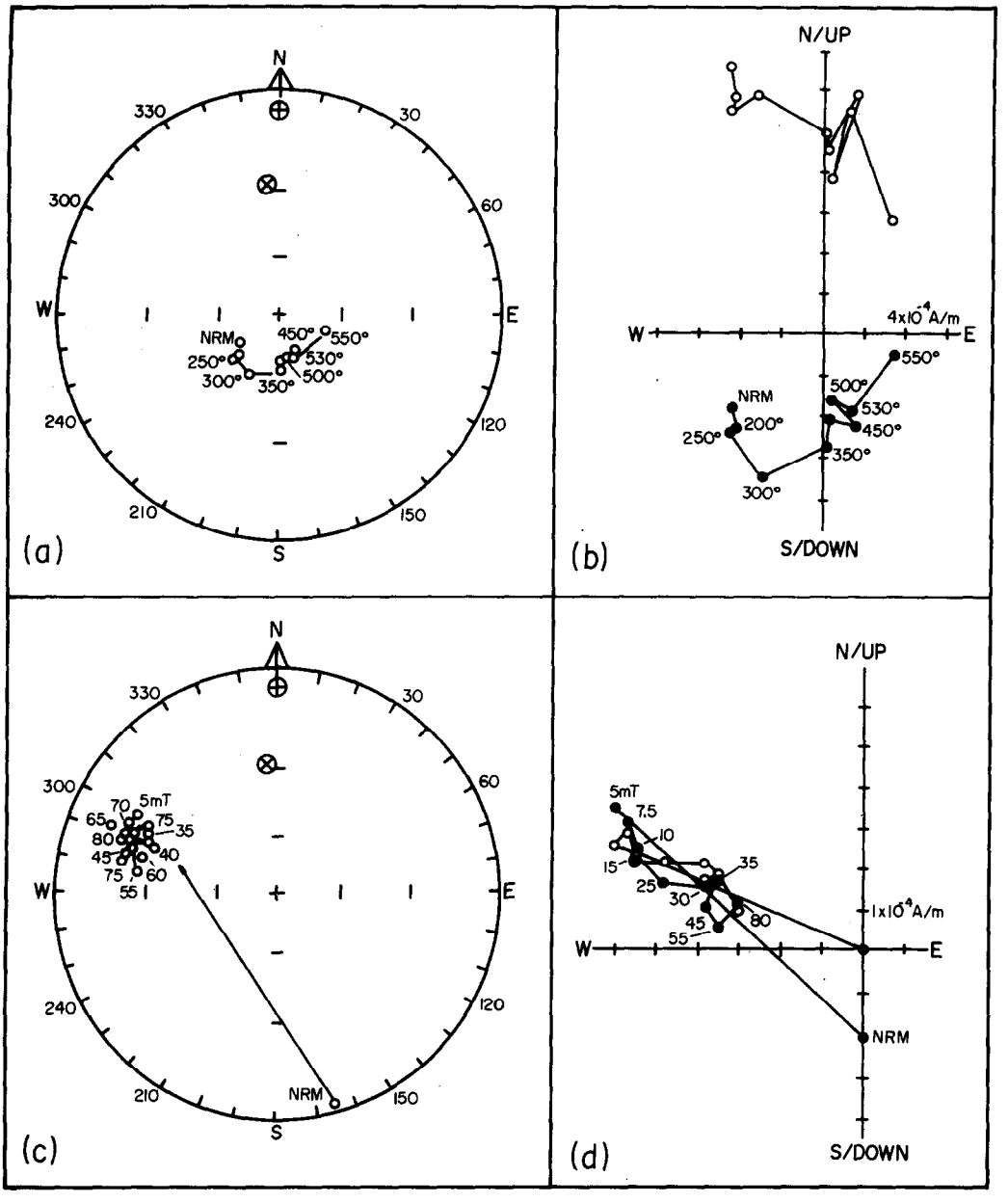


Figure 6. Wulff stereographic projection and Zijderveld diagrams of the magnetization directions during treatment for samples (a, b) OG-56-B1, site S16 and (c, d) OG-86-A1, site S18 (metamorphosed dykes). Same conventions as for Fig. 2.

PALAEOMAGNETIC RESULTS

Ogooué Series

The 27 amphibolites from the Ogooué Series yielded natural remnant magnetization (NRM) intensities ranging from 10^{-4} A m^{-1} to values as high as 20 A m^{-1} and volumetric susceptibilities in the range $6x10^{-4}$-13 (SI units). The magnetization of most samples behaved unstably during AF and thermal treatment (Fig. 2) and it was not possible to resolve their magnetization components. This was the case for sites S7 (OG-1 to 3), S8 (OG-4 to 6), S10 (OG-10 to 12), S12 (OG-16 to 18), S13 (OG-50 to 52) and S14 (OG-99 to 101) (Table 1). Magnetic vector components were obtained from three sites of the Ogooué Series (sites S9, S11 and S15). Specimens from site S9 (OG-7 and OG-9) yielded a southeastern direction with steep upward inclination after AF treatment (OG-9-B1; Fig. 3a, d).

Several specimens of samples from sites S11 (Fig. 3b, d; OG-14-B1) and S15 (Fig. 3c, d; OG-103-C1) yielded a northeastern direction with steep downward inclination after AF and thermal treatments. Specimens from samples OG-15 and OG-102 behaved unstably and only for specimen OG-15-C1 was it possible to isolate the northeastern direction mentioned above.

Archean amphibolites

The nine Archean amphibolites yielded NRM

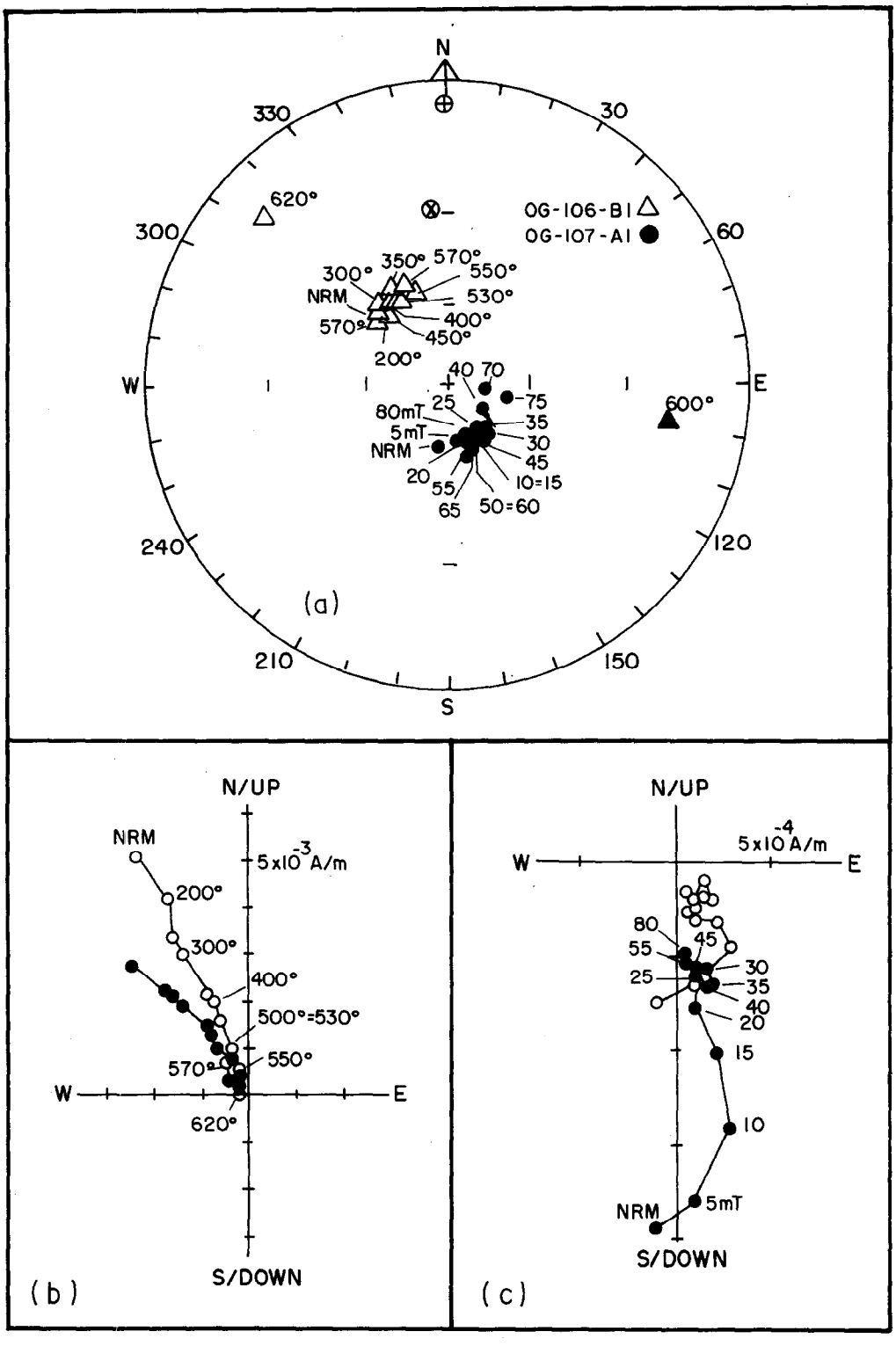


Figure 7. Wulff stereographic projection and Zijderveld diagrams of the magnetization direction during treatment for samples from site S20 (metamorphosed dyke). Same conventions as for Fig. 2.

intensities ranging from 10^{-5} to 10^{-2} A m^{-1}, and susceptibilies in the range $3.8x10^{-4}$-$1.1x10^{-2}$ (SI units). The magnetizationof the majority of samples behaved unstably during AF and thermal treatments. For sample OG-63 (the least weathered sample from site S4), a westerly direction with relatively steep upward inclination could be isolated (Fig. 4a, b). For samples OG-79 and OG-80 from site S5, a northwestern direction (not well defined for all samples) with moderate to steep downward inclinations could be isolated (Fig. 4c, d). The magnetization intensities of these samples (around 10^{-5}A m^{-1}) were close to the magnetometer sensitivity limit, which rendered the definition of magnetic components difficult (see Fig. 4d).

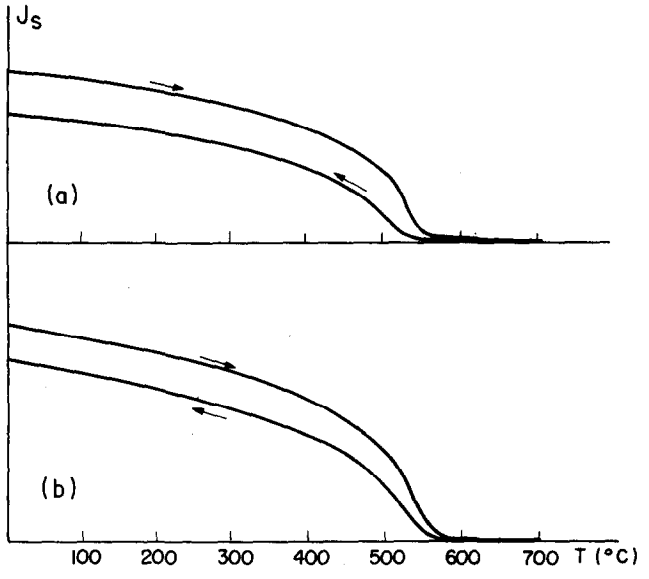


Figure 8. Examples of thermomagnetic curves obtained for the unmetamorphosed dykes. (a) sample OG-70 (site S21), (b) sample OG-72 (site S22).

Basement granulites

The five basement granulites sampled from two sites (S1 and S2) yielded NRM intensities ranging from 1 to 10 A m^{-1} and susceptibilities in the range $6.3x10^{-2}$-0.12 (SI units). A stable east-northeast direction with steep downward inclination could be isolated for all samples after AF and thermal treatment (Fig. 5).

Dolerite dykes

The dolerite dykes were analysed palaeomagnetically and geochemically at the University of Trieste by Enzo Piccirillo. Thin sections of the dykes showed that only five of them were unmetamorphosed, the remainder are metamorphosed to amphibolite facies. The unmetamorphosed dykes are geochemically similar to those from Salvador and Ilhéus-Olivença in Brazil. These dykes correspond to lati-basalt, andesi-basalt and tholeiite rocks (E. Piccirillo, *pers. comm.*, 1994).

Metamorphosed dykes

The metamorphic dykes show NRM intensities ranging from 10^{-4} to 10^{-2} A m^{-1} and susceptibilities in the range $4x10^{-5}$-10^{-4} (CGS units). Several samples behaved unstably during AF and thermal treatments. This was the case for samples from sites S17 and S19. Relatively stable directions could be isolated for samples from sites S16, S18 and S20, although no consistency between site mean directions was obtained. Figures 6 and 7 show examples of the magnetic behavior yielded for each of these sites. Directions with approximately opposite polarities were isolated for two samples from the same dyke (Fig. 7).

Unmetamorphosed dykes

The unmetamorphosed dykes have NRM intensities ranging from 1 to 30 A m^{-1} and susceptibilities in the range $1.9x10^{-2}$-$4.7x10^{-2}$ (SI units). The AF treatment provided better results than the thermal treatment and the majority of the samples showed multicomponent behavior. Thermomagnetic plots are typical of rocks with poor-Ti titanomagnetites (Fig. 8) with Curie temperatures between 544°C and 582°C.

Four dykes yielded either a western direction with moderate to high upward inclination or an eastern direction with moderate to high downward inclination (Figs 9 and 10). Although other directions with higher coercivities were also present, they have not yielded consistent within-site results. Samples from the 5° unmetamorphosed dykes (site S24) yielded multicomponent behavior after AF treatment (Fig. 11). However, none of these directions were found in samples from other sites.

DISCUSSION

The palaeomagnetic results obtained in this study are summarized in Table 1. Amphibolites from the Ogooué Series were metamorphosed during the development of the Ogooué mobile belt (Ledru *et al.*, 1989) during the Eburnean cycle and are Palaeoproterozoic in age. A Rb/Sr regression line based on five muscovites from pegmatites cutting the Ogooué system in an area defined by 11°24'-11°30'E and 0°45'-1°12'S yielded an age of *ca* 1990 Ma (I.R. of 0.700), which post-dates the metamorphism of the series (Vachette, 1964). The Archean amphibolites and granulites are geographically situated inside or at the boundary of the craton next to the Ogooué belt and were probably affected by the Eburnean cycle. At this cratonic margin, the Francevillian sediments (Booué basin) have undergone a deformation phase accompanied by low-grade metamorphism (Ledru *et al.*, 1989).

Petrographical analyses of the dolerite dykes showed that half of the dykes were metamorphosed to amphibolite facies. The geographical proximity of both metamorphosed and unmetamorphosed dykes (Fig. 1) suggests different origins for them and, consequently, different ages for these two groups of dykes. The metamorphic dykes are older than the unmetamorphosed dykes and a possible age for the metamorphism of these dykes would be the Palaeoproterozoic Eburnean cycle.

Most of the amphibolites showed unstable behavior during AF and thermal treatments. However, it was possible to isolate stable directions for amphibolites from three sites of the Ogooué Series. Also, some amphibolites from the Archean basement (two sites) and some amphibolite dykes (3 sites) yielded

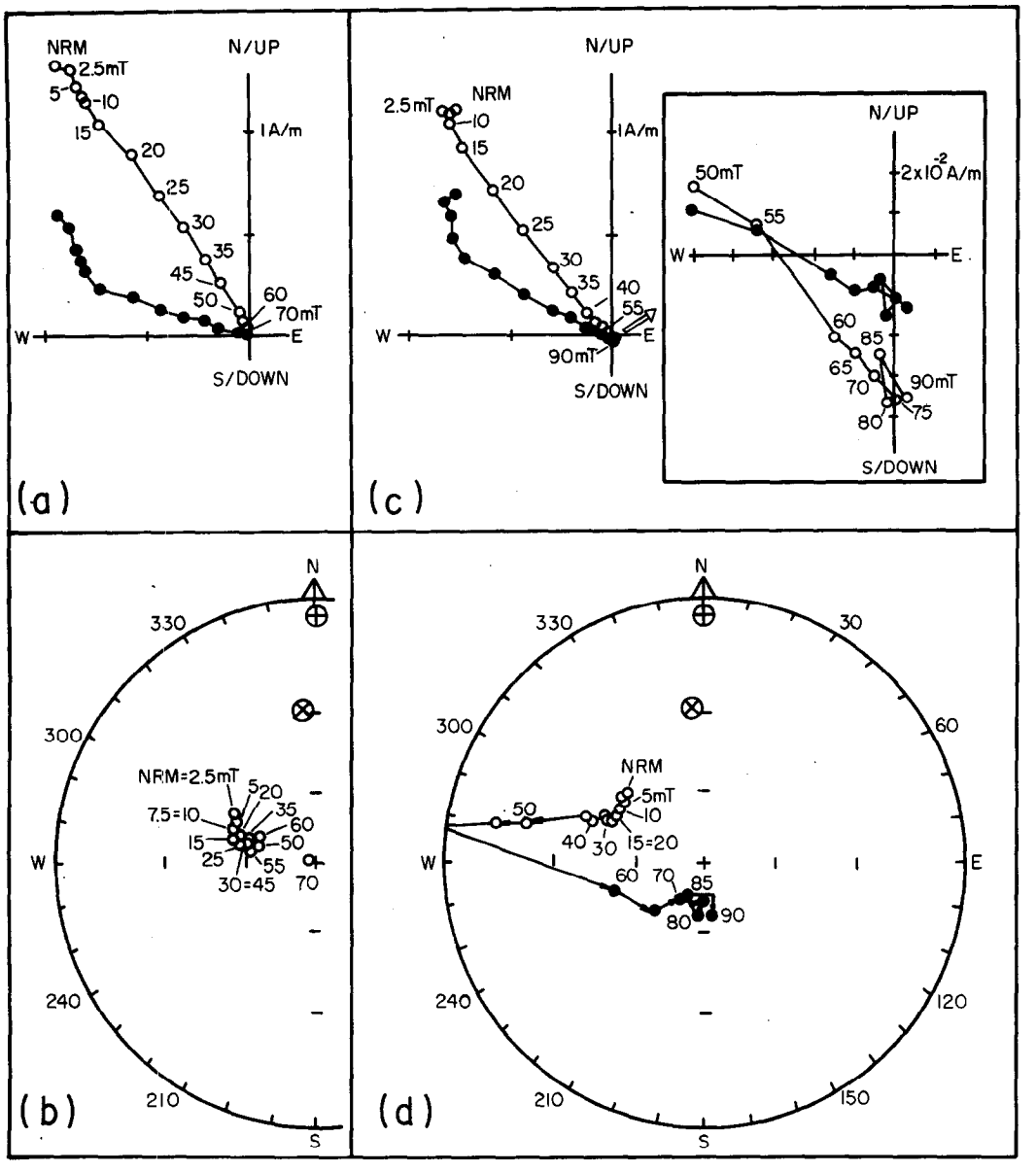


Figure 9. Zijderveld diagrams and wulff stereographic projection for samples (a, b) OG-65-A1 and (c, d) OG-69-A1 from site S21 (unmetamorphosed dyke). Same conventions as for Fig. 2.

stable directions (Table 1). As described earlier, the granulites from two sites (S1 and S2; Table 1) from the Archean basement presented very good stable directions. The site mean directions calculated for these amphibolites and granulites are shown in Fig. 12a. Virtual geomagnetic poles (VGPs) were calculated from these site mean directions (Table 1) and are shown in Fig. 13 (open symbols). The best group of site mean directions is obtained with sites S1, S2, S11 and S15 (Fig. 12a) where two sites (S11 and S15) correspond to amphibolites from the Ogooué Series (Palaeoproterozoic) and two sites (S1 and S2) are basement granulites. Although these directions were isolated for rocks of different lithologies and no radiometric ages are presently available for them, the results are very similar to those obtained for granulites from the Archean Jequié granulitic complex (Brazil), which was affected by the Transamazonian cycle (equivalent to the Eburnean cycle). The calculated mean direction obtained for these four sites of granulites and amphibolites from Gabon, Dm=54.0°; Im=68.6° (κ=27.5; α_{95} =17.8°), yielded a palaeomagnetic pole located at 44.3° E; 19.0° N (K=10.5; A_{95}=29.8°; pole SOG in Fig. 13). This pole, in a pre-drift diagram, is very close to a preliminary pole calculated for granulites from the Jequié complex (JC, Fig. 13), which was dated by the $^{40}Ar/^{39}Ar$ method yielding concordant ages of 1994±16 Ma (biotite) and 2019±25 Ma (hornblende) for two sites (Sletten, 1989).

The S9 VGP (Fig. 13) from another amphibolite site from the Ogooué Series is closer to the pole obtained for Lower Proterozoic dykes situated next to the town of Uauá in Bahia State, São Francisco

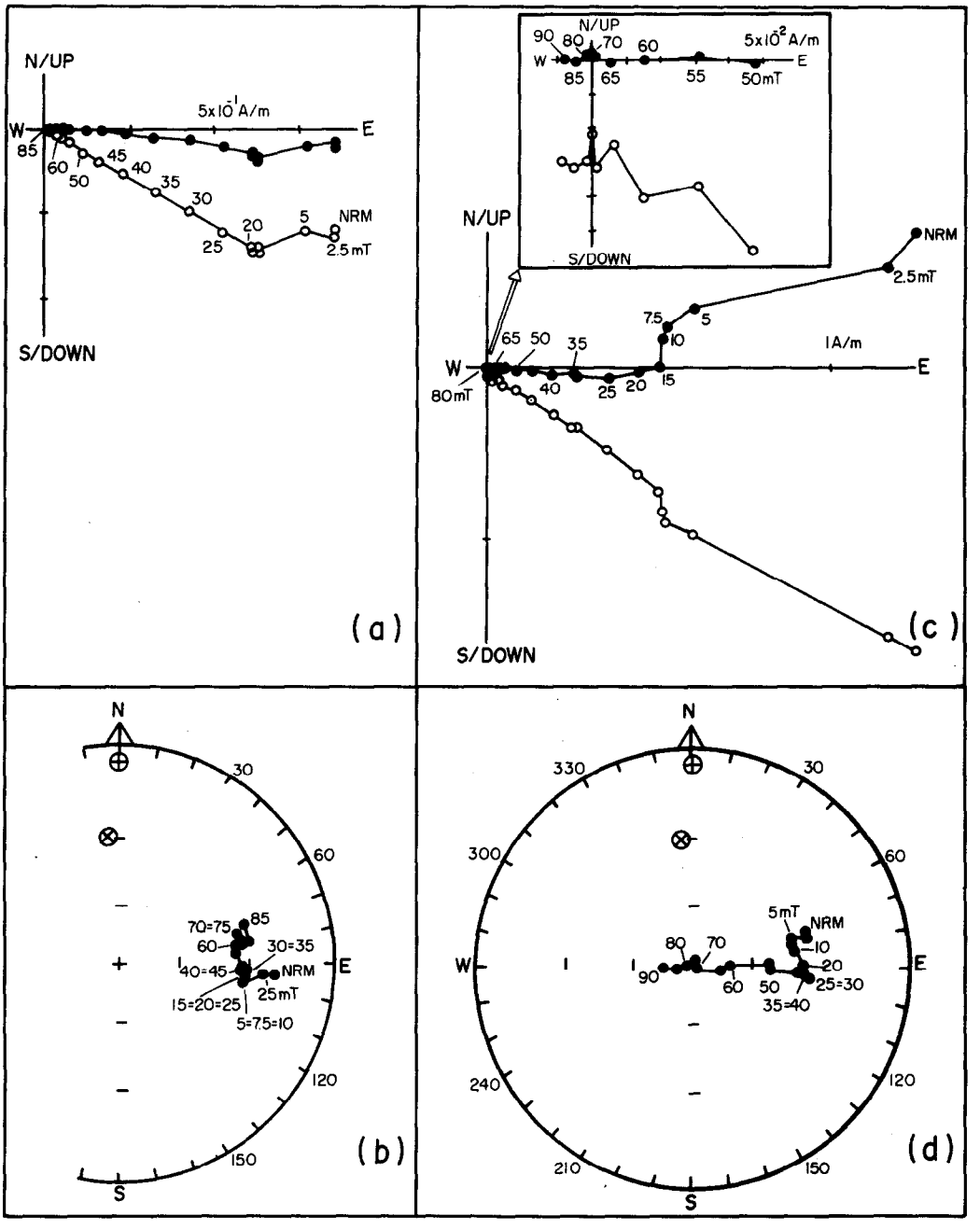


Figure 10. Zijderveld diagrams and wulff stereographic projection for samples (a, b) OG-71-A1 and (c, d) OG-72-A1 from site S22 (unmetamorphosed dyke). Same conventions as for Fig. 2.

craton (d'Agrella-Filho and Pacca, 1991). Also, another two sites yielded VGPs [S5 (basement amphibolites) and S16 (metamorphosed dyke)] which were close to the Uauá pole.

The post-orogenic uplift and slow cooling of the Eburnean cycle may perhaps be responsible for the pole spread observed in Fig. 13, although these poles may also represent later overprints. However, the same characteristic pole spread was also found for granulites and amphibolites from the Salvador area (eastern São Francisco craton, Brazil), which was affected by the equivalent Transamazonian cycle (d'Agrella-Filho, 1992).

Although stable directions could be obtained for the five unmetamorphosed dolerite dykes, multicomponents were present in most samples, as could be observed in the examples presented before. However, only one of these components was carried by samples from four dykes (sites S21, S22, S23 and S25; Table 1, Fig. 12b). A mean direction of magnetization was calculated; Dm=262°, Im= -49° (κ=13; α_{95}=27°), which yielded the palaeomagnetic pole located at 72° E; 8° N (K=9; A_{95}=33°). This palaeomagnetic pole is located far from the palaeomagnetic poles determined for Salvador and Ilhéus-Olivença dykes from Brazil (Fig. 14).

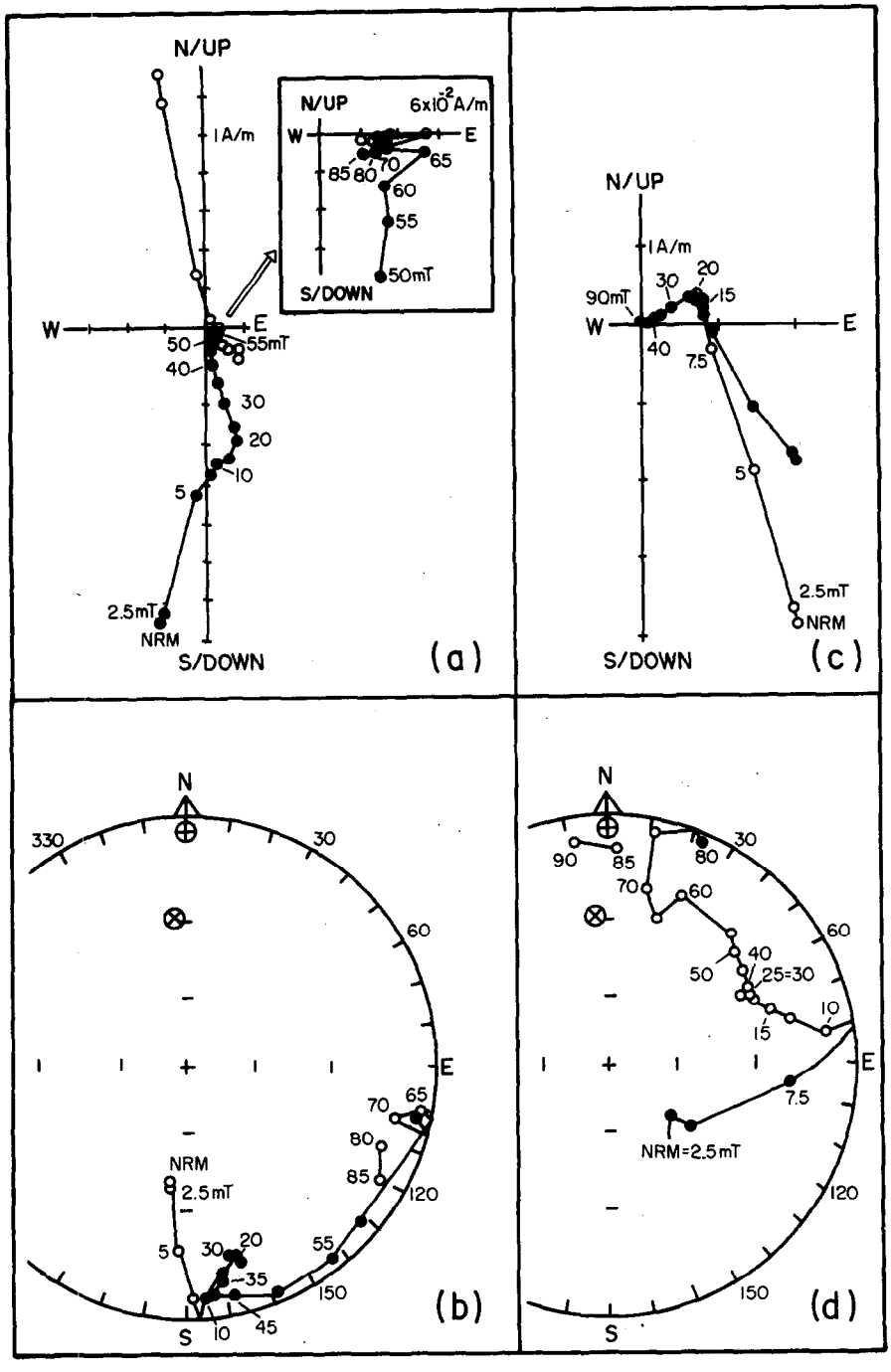


Figure 11. Zijderveld diagrams and wulff stereographic projection for samples (a, b) OG-87-A1 and (c, d) OG-89-A1 from site S24 (unmetamorphosed dyke). Same conventions as for Fig. 2.

Three possible explanations for this inconsistency are:

i) this component of magnetization is of secondary origin - this explanation is coherent with the fact that other directions, although inconsistent, are identified in many samples of these dykes;

ii) these dykes did not correspond to those from Salvador and Ihéus-Olivença, having different ages;

iii) The difference between the poles corresponds to relative movements between the two areas.

CONCLUSIONS

Lower Proterozoic amphibolites from the Ogooué Series, amphibolites and granulites from the Archean basement and dolerite dykes from Gabon (Congo craton) were analysed palaeomagnetically. Although a complicated response to AF and thermal treatments was obtained for these rocks, with many samples showing unstable behavior, it was possible to draw some conclusions about these results.

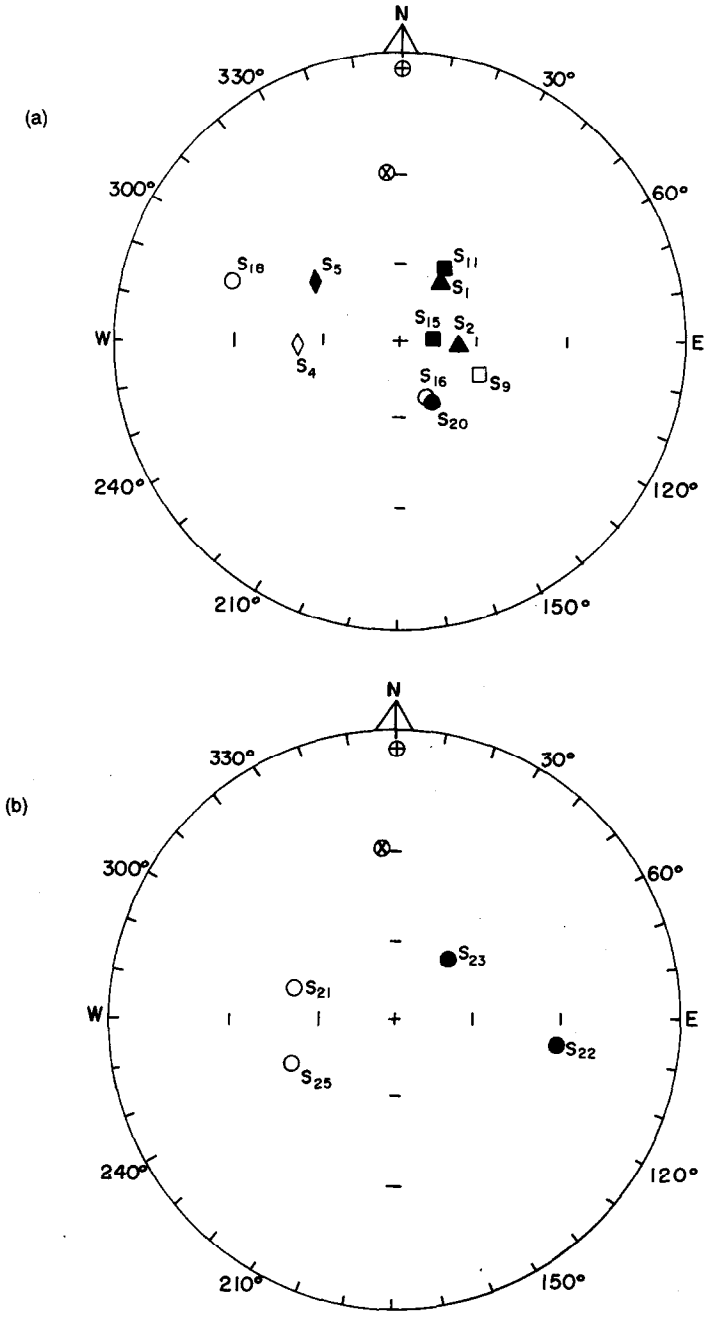


Figure 12. Site mean directions. (a) amphibolites from the Ogooué Series (squares); Archean amphibolites (diamonds); basement granulites (triangles); metamorphic dykes (circles). (b) unmetamorphosed dolerite dykes. Wulff stereographic projection. Same conventions as for Fig. 2.

The analysis showed that granulites present better response to AF and thermal treatments than amphibolites. The same conclusion was obtained analysing amphibolites and granulites from the São Francisco craton.

Characteristic directions were isolated for some of the amphibolite sites and for the two basement granulites sites. Although some (or all) of these directions could represent later overprints, a palaeomagnetic pole (SOG in Fig. 13) determined for the best group of mean site directions (two sites of amphibolites from the Ogooué Series and two sites of granulites) is located very close to the pole obtained for granulites from the Jequié complex (São Francisco craton), probably of 2.0 Ga (pole JC in Fig. 13), suggesting that these areas could belong to the same unit at this time.

The unmetamorphosed Neoproterozoic dolerite dykes yielded a common characteristic direction for four of these dykes. The palaeomagnetic pole (with a high value of α_{95}) calculated for these characteristic directions is very different from those determined for the dyke swarms from Salvador and Ilhéus-Olivença in Brazil (Fig. 14).

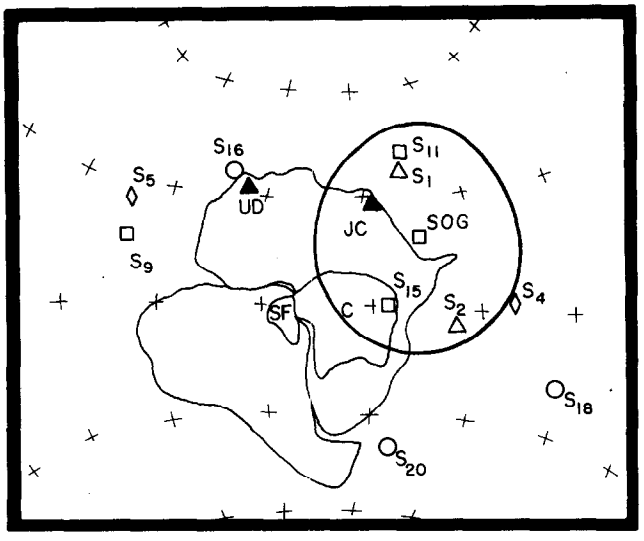


Figure 13. Virtual geomagnetic poles calculated for the analysed sites. Symbols are as in Fig. 11. SOG is the mean of the following VGPs: S1, S2, S11 and S15. Poles UD (Uauá dykes) and JC (Jequié complex) are from the São Francisco craton. SF- São Francisco craton. C- Congo craton. Africa in its present position.

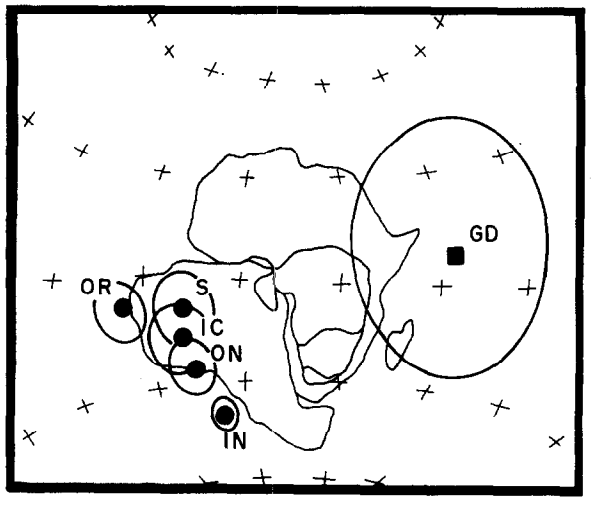


Figure 14. Comparison of the pole (GD) calculated for the site mean directions of Fig. 12b (Gabon dolerite dykes) with those obtained for Ilhéus-Olivença (OR, IC, ON, IN) and Salvador (S) dykes from the São Francisco craton (D'Agrella-Filho *et al*, 1990; D'Agrella-Filho, 1992). Africa in its present position.

Three alternatives to explain this difference are: the dykes have different ages, or the characteristic direction found for the dolerite dykes represents a secondary component, or this difference represents relative movements between these areas (less probable). $^{40}Ar/^{39}Ar$ datings of these rocks are in progress (Paul Renne from the Institute of Human Origins, Berkeley, USA), which may help to elucidate this problem.

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